

Energy Ledgers for Forced Harmonic ODEs

Kirchhoff power balance and an unknown component with $X = LRC$

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Abstract

We add an unknown term Xq''' to a forced harmonic LRC equation and determine whether it supplies or absorbs energy by integrating its signed power. Exact trajectories exhibit both roles. For $X = LRC$, coefficient choices scale the transfer on a fixed trajectory; its direction follows from the derivative product. Brief hammer and rotary applications reuse these rules. The unknown component's remaining energy account is left open.

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Latest on GitHub: <https://github.com/hobnilre/physics-ode-energy>

1 Kirchhoff's voltage law and the missing accounts

Use the [LRC template](#), Section 2, with the [series connection convention](#), Section 1:

$$Lq'' + Rq' + \frac{q}{C} = f(t), \quad L > 0, \quad C > 0, \quad R \geq 0. \quad (1)$$

Here q is charge, $i = q'$ current and f the applied forcing voltage; L, R, C are inductance, resistance and capacitance. Coefficients are constant, trajectories smooth, and primes denote time derivatives. Multiplying by i and putting the source first gives

$$f(t)i - Lq''i - Rq'i - \frac{qi}{C} = 0. \quad (2)$$

Every term is voltage times current. The input contributes $+fi$ and the resistor contributes $-Ri^2$: the single-loop power identity of [Tellegen \(1952\)](#).

The power sum is zero. To relate it to constant total energy, include the heat recipient and the forcing source or sink:

$$\begin{cases} 1 : & \dots + Rq'i + \dots = 0, \\ 2 : & f(t)i - Lq''i - Rq'i - qi/C = 0, \\ 3 : & -f(t)i + \dots = 0. \end{cases} \quad (3)$$

Lines 1 and 3 belong to the heat recipient and the forcing source or sink. Their entries $+Ri^2$ and $-fi$ oppose the transfers in line 2; the remaining terms are unspecified. If $fi < 0$, the forcing source receives energy.

Cancellation identifies the transfers. Constant total energy also requires the omitted energy rates and reservoir laws. Those accounts remain open; a prescribed $f(t)$ alone does not specify the source's available energy.

2 Does the unknown term Xq''' supply or absorb power?

2.1 The fourth line and the direction of transfer

Add one fictional component term, with constant $X > 0$:

$$Xq''' + Lq'' + Rq' + \frac{q}{C} = f(t). \quad (4)$$

Its physical identity is unknown. Treat Xq''' as its voltage contribution. The power equation and its matching entries become

$$\begin{cases} 1 : & \dots + Rq'i + \dots = 0, \\ 2 : & f(t)i - Xq'''i - Lq''i - Rq'i - qi/C = 0, \\ 3 : & -f(t)i + \dots = 0, \\ 4 : & \dots + Xq'''i + \dots = 0. \end{cases} \quad (5)$$

The fourth line carries the opposite of the new contribution in line 2. Define the two signed powers explicitly:

$$P_X = Xq'''i \quad (\text{received by } X), \quad P_{X \rightarrow 2} = -Xq'''i \quad (\text{contribution to line 2}). \quad (6)$$

Table 1. Direction follows from the derivative product because $X > 0$.

Sign of $q'''i$	P_X	$P_{X \rightarrow 2}$	Role of X
Negative	Negative	Positive	Supplier
Positive	Positive	Negative	Absorber
Zero	Zero	Zero	No instantaneous transfer

Energy transfer follows from the signed power integral:

$$W_X(t_0, t_1) = \int_{t_0}^{t_1} Xq'''(t)i(t) dt, \quad W_{X \rightarrow 2} = -W_X. \quad (7)$$

Positive W_X is net energy received by X ; negative W_X is net energy supplied by it. This specifies the required transfer while leaving the remaining terms of line 4 open.

2.2 Two numerical examples from exact trajectories

Take the illustrative values $L = 1 \text{ H}$, $R = 1 \Omega$, $C = 1 \text{ F}$, and $X = 1 \text{ H s}$. Put $u = t/(1 \text{ s})$. In each example choose $q(t)$ and obtain the required $f(t)$ by substitution into (4).

Supplier. Choose

$$q(t) = (1 \text{ C}) \sin u, \quad i(t) = (1 \text{ A}) \cos u, \quad q'''(t) = -(1 \text{ A/s}^2) \cos u. \quad (8)$$

The inductive and capacitive voltages cancel, as do the resistive and X voltages, so the required forcing is $f(t) = 0$. The two powers are

$$P_X = -(1 \text{ W}) \cos^2 u, \quad P_{X \rightarrow 2} = +(1 \text{ W}) \cos^2 u. \quad (9)$$

At $t = 0$, X contributes $+1$ W to line 2. Over the exact period $T = 2\pi$ s,

$$W_X = -(1 \text{ J}) \int_0^{2\pi} \cos^2 u \, du = -\pi \text{ J}. \quad (10)$$

Thus X supplies π J during the cycle. Here $P_X = -Ri^2$ pointwise, so the required transfer pays for the resistor's absorption. The equation identifies the required debit in the unknown account; it does not specify what physically supplies that energy.

Absorber. On $0 \leq t \leq 1$ s, choose

$$q(t) = (1 \text{ C}) \frac{u^3}{6}, \quad i(t) = (1 \text{ A}) \frac{u^2}{2}, \quad q'''(t) = 1 \text{ A/s}^2. \quad (11)$$

Direct substitution gives

$$f(t) = (1 \text{ V}) \left(1 + u + \frac{u^2}{2} + \frac{u^3}{6} \right). \quad (12)$$

Now $P_X = (1 \text{ W})u^2/2 > 0$ for $t > 0$, so X absorbs power. At $t = 1$ s, its contribution to line 2 is $P_{X \rightarrow 2} = -1/2$ W. Its received energy is

$$W_X = (1 \text{ J}) \int_0^1 \frac{u^2}{2} \, du = \frac{1}{6} \text{ J}. \quad (13)$$

The same positive X therefore supplies energy on the sinusoid and absorbs energy on the polynomial trajectory. Both examples are exact substitutions and evaluated power integrals.

3 Choosing L , R and C when $X = LRC$

3.1 Units and what the coefficient determines

For $(a, b, c) = (L, R, 1/C)$, the family and staircase in *ODE Coefficient Synthesis*, Sections 1–3, give

$$X = A_3 = A_{-1,3} = \frac{ab}{c} = LRC > 0. \quad (14)$$

Here $R > 0$. This is the chosen coefficient; the component's identity remains unknown. Since RC has units of time, $[X] = \text{H s}$ and $[Xq'''] = \text{V}$. Its contribution to line 2 is

$$P_{X \rightarrow 2} = -LRC q''' i. \quad (15)$$

3.2 Maximizing the supplying or absorbing contribution

First hold the trajectory $q(t)$ fixed, including the time scale and amplitude. Then the derivative product $q''' i$ is fixed at each instant. Equation (15) gives the two design cases directly:

Table 2. At fixed trajectory, both magnitudes increase with the product LRC .

Objective at an instant	Required trajectory sign	Magnitude to maximize
Supply power to line 2	$q''' i < 0$	$LRC(-q''' i)$
Absorb power from line 2	$q''' i > 0$	$LRC(q''' i)$

Increasing any of L , R , or C strengthens the existing direction of transfer. Positive component choices

cannot reverse that direction for the same trajectory. With independent positive bounds on the three components, the largest product uses all three upper bounds. Without such bounds there is no finite maximum. The signed work in (7) scales in the same way when the trajectory and interval are fixed.

For the sinusoidal family $q = Q \sin(\omega t)$, where $Q > 0$ is charge amplitude and $\omega > 0$ is angular frequency,

$$q''' = -\omega^2 i, \quad P_{X \rightarrow 2} = LRC \omega^2 i^2 \geq 0. \quad (16)$$

Every positive choice of L, R, C therefore makes X a supplier on this trajectory, except at current zeros. Absorption requires a trajectory with $q''' i > 0$, such as the polynomial in (11); changing positive coefficients alone cannot produce it on the sinusoid.

If “effect” instead means magnitude relative to the resistor’s absorption, then, at $i \neq 0$,

$$\frac{|P_{X \rightarrow 2}|}{Ri^2} = LC \left| \frac{q'''}{i} \right|. \quad (17)$$

At fixed trajectory, larger L or C increases this ratio. Larger R increases both powers equally and leaves the ratio unchanged. On a sinusoid the ratio is $LC\omega^2$.

3.3 Numerical comparison in both roles

Use the two trajectories from Part 2. Compare the supplier at $t = 0$ and the absorber at $t = 1$ s. For each component choice, recompute $f(t)$ from (4) so that the specified trajectory is retained.

Table 3. Exact signed contributions $P_{X \rightarrow 2}$ of the unknown term.

(L, R, C) in (H, Ω , F)	X (H s)	Supplier (W)	Absorber (W)
(1, 1, 1)	1	+1	−1/2
(1, 2, 1)	2	+2	−1
(2, 2, 2)	8	+8	−4

With independent bounds $1 \leq L/(1 \text{ H}) \leq 2$, $1 \leq R/(1 \Omega) \leq 2$ and $1 \leq C/(1 \text{ F}) \leq 2$, the last row maximizes both directional magnitudes. Part 2’s integrals then give $8\pi \text{ J}$ supplied over the sinusoidal cycle and $4/3 \text{ J}$ absorbed over the polynomial interval.

These comparisons hold the motion fixed and adjust the forcing accordingly. If $f(t)$ and the initial conditions are held fixed instead, changing L, R, C also changes $q(t)$; maximizing the product alone then does not determine the maximum effect. The coefficient relation specifies the power required of X on a chosen motion. Its physical identity and the remaining terms of line 4 are still open.

4 Hammer and nail: applying Parts 2 and 3

Let x be hammer-face and nail-head movement during smooth contact, $v = x'$, and $m, d, k > 0$ constant mass, resistance and stiffness. The domain map $(L, R, C, q, i) \mapsto (m, d, 1/k, x, v)$ gives

$$X_h x''' + mx'' + dx' + kx = F(t), \quad X_h = \frac{md}{k}. \quad (18)$$

Here F is external driving force. Part 2 gives received power $X_h x''' v$: its sign determines supply or absorption. Part 3 gives the fixed-motion scaling through md/k , with F recomputed.

For one exact blow, set $F = 0$, $x(0) = 0$, $v(0) = v_0 > 0$ and $x''(0) = 0$. With $\omega = \sqrt{k/m}$, the solution up to the first stop is

$$x(t) = \frac{v_0}{\omega} \sin(\omega t), \quad 0 \leq t \leq t_* := \frac{\pi}{2\omega}. \quad (19)$$

Indeed, $mx'' + kx = 0$ and $X_h x''' + dv = 0$. Applying (7) gives

$$W_{X_h} = \int_0^{t_*} X_h x''' v dt = - \int_0^{t_*} dv^2 dt = - \frac{\pi dv_0^2}{4\omega} = -W_d. \quad (20)$$

The component supplies the resistance work from its unspecified account. Movement x includes elastic support motion; permanent nail depth is undetermined.

5 Rotary impact driver: work on a prescribed motion

Let θ be output angle, $\Omega = \theta'$, and $J, d, k > 0$ constant inertia, damping and torsional stiffness. Prescribed hammer torque is τ_h ; J excludes the hammer. Substituting $(L, R, C) \mapsto (J, d, 1/k)$ gives

$$X_r \theta''' + J \theta'' + d \theta' + k \theta = \tau_h, \quad X_r = \frac{Jd}{k}. \quad (21)$$

At identical motion and coefficients, let τ_X and τ_0 be the required torques with and without the unknown term. Part 2's work rule gives

$$\Delta W_h = \int_{t_0}^{t_1} (\tau_X - \tau_0) \Omega dt = \int_{t_0}^{t_1} X_r \theta''' \Omega dt = W_{X_r}. \quad (22)$$

Thus supplied work reduces the required hammer input on that motion.

Take $\theta = \Theta \sin(\omega t)$, with $\Theta, \omega > 0$, on $0 \leq t \leq T := \pi/(2\omega)$. Since $\theta''' = -\omega^2 \Omega$,

$$\begin{aligned} W_{X_r} &= \int_0^T X_r \theta''' \Omega dt = - \frac{\pi X_r \Theta^2 \omega^3}{4}, \\ W_d &= \int_0^T d \Omega^2 dt = \frac{\pi d \Theta^2 \omega}{4}, \quad \frac{-W_{X_r}}{W_d} = \frac{J \omega^2}{k}. \end{aligned} \quad (23)$$

The ratio is Part 3's $LC\omega^2$. On this motion, an ordinary model requires the same torque with damping

$$d_{\text{eff}} = d - X_r \omega^2 = d \left(1 - \frac{J \omega^2}{k} \right) \geq 0. \quad (24)$$

For example, $J\omega^2/k = 1/10$ gives $d_{\text{eff}} = 9d/10$ and $W_{X_r} = -W_d/10$. The unknown component supplies work that the reduced damping avoids dissipating.

The forcing is recomputed, as in Part 3; matching this motion does not establish equal responses to a fixed drive. Hammer preparation and the account of X_r remain outside the output-work comparison.

References

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3. H. Nilre and B. C. Herlin (2026). [ODE Coefficient Synthesis](#).
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